HYPERBOLIC EXTENSIONS OF GLOT: A STAGE-WISE STUDY OF GRAPH-BASED SENTENCE POOLING

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ABSTRACT

Graph-based pooling builds a latent token graph over a frozen language model and refines token states with a GNN before aggregation. GLOT (Mantri et al., 2026) is a strong recent instance, matching much larger poolers with $20\times$ fewer trainable parameters while keeping the backbone frozen. We ask whether hyperbolic geometry, which embeds hierarchical structure more efficiently than Euclidean space, can extend it, and we study three stages independently: a Poincaré token graph (**A**), a gyro-midpoint readout (**B**), and Möbius message passing (**C**). Across 990 runs with 15 paired seeds and an equal search budget per stage over GLOT's own grid, **Stage C is the strongest**: $+1.42$ MCC on CoLA with $13/15$ seeds positive, significant on both a paired t-test and an exact sign test, and A+C is the best arm on STS-B. **Stage B is significantly harmful** on both an encoder and a decoder backbone, surviving Bonferroni. "Hyperbolic pooling" should therefore not be treated as one design choice: its readout and its message passing move in opposite directions, and conflating them hides both.

Two findings about GLOT itself follow from the same experiments, and both are in its favour. It is *better than its own paper reports*: at the authors' seed and over their published grid we exceed the reported CoLA by 3.03 MCC and RTE by 0.36, because the published numbers are single-seed maxima over a search rather than means over seeds. And it is markedly *robust to how its token graph is built*: realised edge density ranges from 0.10 on BERT to 0.99 on RoBERTa under the identical threshold, yet accuracy is unchanged, and GLOT beats mean, max and `[CLS]` pooling by 14–35 points in every seed of the authors' noise diagnostic. That robustness is practically convenient — the method ports to a new backbone without re-tuning the graph — and it also indicates that the gain is carried by the trained readout rather than by the particular graph, which is where we would direct further work on this family.

1 INTRODUCTION

Reducing a language model's token states to a single vector is a prerequisite for most sentence-level tasks. Standard poolers — mean, max, `[CLS]` — treat tokens as an unordered set. A recent line of work instead frames pooling as *relational learning followed by aggregation*: build a latent graph over tokens, refine it with a GNN, then aggregate. GLOT (Mantri et al., 2026) is a strong instance, reporting competitive GLUE and MTEB results with $20\times$ fewer trainable parameters than comparable methods while keeping the backbone frozen. Our experiments confirm the premise it rests on: on the authors' own noise diagnostic, GLOT beats every set-based pooler by 14–35 points in every seed we ran.

This paper asks what *else* that architecture can support. Token states form an approximately hierarchical structure, and hyperbolic space embeds hierarchies with far less distortion than Euclidean space, so it is a natural candidate for each of GLOT's three components. We therefore build hyperbolic counterparts of all three — the graph, the readout, and the message passing — and measure them separately rather than as a bundle. That separation turns out to matter: two of the three move in opposite directions, and a study that treated "hyperbolic pooling" as one switch would have reported their average and concluded nothing.

[figure placeholder — see caption]

Figure 1: GLOT's pooling pipeline and the three modifications we test. Token states from a frozen backbone are thresholded into a graph (Eq. 1), refined by a GNN, and aggregated by a learned readout. Stage A changes the graph construction, Stage B the readout, Stage C the message passing; the `no_graph` control removes the edges while leaving every parameter in place. The lower panel shows the calibration failure: a fixed τ produces very different densities on different backbones.

Contributions.

1. **A stage-wise hyperbolic extension** (§5). Under an equal-budget search over GLOT's own grid with 15 paired seeds, hyperbolic message passing (Stage C) gives $+1.42$ MCC on CoLA with $13/15$ seeds positive, significant on both tests, and A+C is the best arm on STS-B. The hyperbolic readout (Stage B) is significantly harmful on both an encoder and a decoder backbone. Reporting "hyperbolic" as a single factor would have cancelled these out.
2. **GLOT is under-reported** (§3). The published numbers fix seed 42 and take the best τ per task on the split they report. At that seed, over that grid, we reach 50.52 MCC on CoLA against a reported 47.49, and 59.57 on RTE against 59.21.
3. **Porting GLOT to a new backbone** (§4). An absolute cosine threshold produces very different graphs on different backbones — density 0.10 on BERT, 0.99 on RoBERTa. Accuracy is nonetheless unchanged, so the method transfers as-is; we show that the two natural "corrections" (density-matching, mean-based norm rescaling) are the interventions that actually cost accuracy, and give the measurements a practitioner needs to avoid them.

2 METHOD AND VARIANTS

Let $H = (x_1, \ldots, x_L)$ be the token states of a frozen backbone. GLOT forms

$$A_{ij} = \mathbf{1}[\cos(x_i, x_j) > \tau], \tag{1}$$

applies K graph-attention layers with jumping-knowledge concatenation, and aggregates the refined states with a learned readout. Only the pooler trains.

We evaluate three orthogonal modifications and all their combinations (Figure 1). **Stage A** replaces Eq. 1 with a threshold on *geodesic* distance in the Poincaré ball after $\exp_0^c(\cdot)$. **Stage B** replaces the readout with a Fréchet-style mean in the ball. **Stage C** performs message passing with Möbius operations. We additionally run a `no_graph` control, implemented by raising τ until essentially no edge survives, so that parameter count and architecture are unchanged and only the relational information is removed.

Protocol. A tuning stage draws configurations at a held-out seed; a confirmation stage re-runs the selection on 15 seeds shared across all arms, so every comparison is paired. We report an exact

Table 1: Left: the same baseline runs summarised three ways. Right: best score at GLOT's own seed (42) over GLOT's own grid, 78 configurations per task. Two of three tasks reproduce and are exceeded.

Estimator (CoLA)	value	Δ	Task	ours	pub.	Δ
Published GLOT	47.49	—	CoLA	**50.52**	47.49	$+3.03$
Mean over 15 seeds *(ours)*	45.37	-2.12	RTE	**59.57**	59.21	$+0.36$
Max over 15 seeds	48.17	$+0.68$	STS-B	82.99	83.86	-0.87
Max over 40 tuning configs	50.16	$+2.67$				

Table 2: Edge density induced by the published grid $\tau \in \{0.1, 0.3, 0.6\}$, with the tenth-percentile token cosine that produces it. Measured directly on 192 CoLA sentences at the **final** layer, over off-diagonal ordered token pairs with padding masked and self-loops excluded; this is a property of the backbone and τ alone, independent of training. Density near 1 means the GNN receives a near-complete graph, and therefore no selective structure.

Backbone	cos p10	τ=0.1	τ=0.3	τ=0.6
TinyLlama-1.1B	0.016	0.762	0.533	**0.073**
ModernBERT-base	0.076	0.866	0.759	0.599
BERT-base	0.098	0.876	0.505	**0.103**
SmolLM2-360M	0.322	0.956	0.908	0.741
RoBERTa-base	**0.693**	1.000	1.000	**0.991**

two-sided sign test alongside the paired t-test and call a result significant only when both agree, with Bonferroni across arms. Every arm receives the same trial budget. Full details, including a cache/RNG confound worth ≈ 5 MCC that dictates run order, are in Appendix A.

3 GLOT IS BETTER THAN REPORTED

Mantri et al. (2026) state that "for all experiments, we used a fixed random seed of 42", and their Table 8 reports the best τ per task on the development split they also report. Every published quality number is therefore a single draw, maximised over a search. We report means over 15 seeds. These are not the same estimator, and the difference explains the apparent gap (Table 1).

Two consequences follow. GLOT performs better than its own paper claims on CoLA and RTE. And the reported τ-sensitivity curve cannot be read as a τ effect: it spans 7.87 MCC on CoLA, while we measure a spread of 5.46 across seeds at *fixed* τ, and both published rows are flat with a single spike that is the value promoted to their Table 1. Because the estimators differ, every comparison below is against our own baseline under identical conditions, never against the published number.

4 PORTING GLOT TO A NEW BACKBONE

Before comparing geometries we had to establish that each backbone receives a well-formed graph. Eq. 1 compares a cosine to an absolute τ, which is meaningful only if token-cosine scale is comparable across models. It is not, and Table 2 quantifies by how much: the same grid GLOT publishes induces very different graphs on five backbones. This section reports that measurement, and then the more useful result — that GLOT tolerates the variation, while the obvious corrections do not.

RoBERTa is the extreme case: its *tenth*-percentile token cosine is 0.693, above the largest τ in the grid, so no searched configuration makes its graph sparse and GLOT on RoBERTa operates on an essentially complete graph. SmolLM2 is borderline (0.741 at the sparsest τ). BERT and TinyLlama sit in the regime the threshold was designed for.

The failure is layer-dependent. ModernBERT reaches 0.599 at the final layer but 0.996 at layer 12, which is the layer our fine-tuning experiments use. Mantri et al. (2026) do not state which

Table 3: RoBERTa-base, $n = 15$ paired seeds, tuned per arm and per task on RoBERTa (40 trials each, 40 distinct configurations each). `paper_tau` pins the published grid and yields the near-complete graph of Table 2; `density_fix` is the quantile correction. The correction is the worst arm on both tasks, and on CoLA its deficit survives Bonferroni. Notably `density_fix` *won* the tuning stage on CoLA (56.77, best of all five arms) before finishing last over 15 seeds — so it was favoured, not handicapped, by selection.

	CoLA (MCC)			STS-B (Spearman)		
Arm	mean	Δ_{base}	p/n	mean	Δ_{base}	p/n
baseline	**54.60**	*ref*	—	83.89	*ref*	—
`paper_tau`	54.46	-0.14	8/7	**84.21**	$\mathbf{+0.32}$	12/3
A	54.37	-0.23	5/10	83.80	-0.09	5/10
`no_graph`	54.32	-0.29	5/10	83.84	-0.05	7/8
`density_fix`	53.19	$\mathbf{-1.41}$	2/13	83.73	-0.15	5/10

layer they pool, and Table 2 shows the answer changes the diagnosis — so "which backbones are affected" cannot be settled from the published description alone. We therefore treat RoBERTa as the one unambiguous case.

GLOT tolerates this; density-matching does not. The obvious repair is to replace the absolute threshold with a quantile criterion, retaining the top q fraction of pairs so that realised density is q by construction. We tested it on RoBERTa under the same equal-budget protocol as §5 — 40 tuning trials per arm on RoBERTa itself, per task, then 15 paired confirmation seeds disjoint from the tuning seed. The result is a useful negative for anyone porting this method (Table 3): GLOT at its published threshold is unharmed by the dense graph, while the correction costs accuracy.

The near-complete graph that Table 2 identifies as degenerate is the *best* arm on STS-B ($+0.32$, $12/3$, sign-test $p = 0.035$) and is statistically tied for best on CoLA. Forcing it to be sparse costs 1.41 MCC ($t = -3.44$, $2/13$, surviving Bonferroni at $\alpha = 0.0125$). Against `paper_tau` directly the correction loses on both tasks (-1.27 CoLA, -0.48 STS-B, the latter at $2/13$ with $p = 0.0074$).

Separating density from scale. We had attributed a $21.71 \rightarrow 27.15$ MCC recovery on ModernBERT/CoLA to a "two-part correction" of density-matching plus median-based norm rescaling, measured at a single seed with the two factors varied together. Table 4 reports the full 2×3 factorial over 5 seeds and four backbone/layer combinations, which separates them. Three things follow.

First, **the choice of rescaling statistic matters far more than whether you rescale**, and this is the only effect in the factorial that is significant on both tests. On ModernBERT layer 12, whose mean/median token-norm ratio is 7.66, *mean*-based (`rms`) rescaling costs 11.03 MCC against no rescaling ($t = -5.47$, $0/5$ seeds positive) — the mean is precisely the statistic the outliers corrupt. Density-matching partially rescues that damage ($+11.57$ within the `rms` condition, $5/0$), which is the only context in which density-matching helps anywhere in this paper.

Second, **median rescaling is directionally positive but not yet established**. It gains 5.96 MCC at absolute τ and 5.32 at $q = 0.05$ on layer 12, and 1.34 at the final layer with $5/0$ seeds positive — but with per-seed standard deviations of 7–10 on layer 12, none of these reaches significance at $n = 5$ ($t = 1.38$, 1.93 and 2.24). We flag a structural limitation of this table: at $n = 5$ the exact two-sided sign test cannot return below $p = 0.0625$, so no contrast here can satisfy the criterion we impose everywhere else. We are extending it to the 15 seeds used in the rest of the paper and report the effect as a lead, not a result.

Third, **density-matching does not transfer**. Its main effect is $+4.58$ on ModernBERT layer 12 but -1.91 at ModernBERT's final layer, -0.25 on BERT and $+0.45$ on RoBERTa under this fixed recipe; none is significant, and under the tuned protocol of Table 3 it is significantly harmful on RoBERTa. We therefore withdraw it as a recommendation.

Table 4: Density × scale factorial, CoLA MCC, mean ± sd over 5 seeds, GLOT's published recipe otherwise unchanged. Columns vary the threshold (absolute, as published, versus density-matched); rows vary token-norm rescaling. *Median* rescaling helps only the backbone/layer with extreme norm outliers; density-matching has no consistent sign; BERT is inert throughout. Note also the variance: ModernBERT layer 12 has a seed sd of 10.16 in the published cell, which is why its single-seed numbers should not be read closely.

Backbone (layer)	Rescaling	$\tau = 0.6$ (pub.)	$q = 0.05$
BERT-base (final) *control*	none	45.25 ±0.97	45.00 ±1.51
	`rms`	45.01 ±1.25	45.03 ±1.18
	`median`	44.98 ±1.00	45.08 ±0.83
ModernBERT-base (L12)	none	18.64 ±10.16	23.22 ±6.17
	`rms`	7.61 ±7.71	19.18 ±2.98
	`median`	24.60 ±7.12	**28.54** ±2.41
ModernBERT-base (final)	none	37.53 ±1.09	35.62 ±6.00
	`rms`	38.64 ±1.48	37.61 ±1.72
	`median`	**38.87** ±1.18	37.57 ±1.79
RoBERTa-base (final)	none	52.39 ±1.98	52.89 ±0.97
	`rms`	52.45 ±1.43	–
	`median`	**52.54** ±1.38	–

Table 5: BERT-base, wide search, $n = 15$ paired seeds. Δ_{base} is against GLOT's cosine baseline, Δ_{ng} against the `no_graph` control; "p/n" counts seeds positive/negative against the baseline. Full statistics in Appendix B.

	CoLA (MCC)			STS-B (Spearman)		
Arm	Δ_{base}	p/n	Δ_{ng}	Δ_{base}	p/n	Δ_{ng}
C	$\mathbf{+1.42}$	13/2	$+0.58$	$+0.20$	12/3	$+0.16$
AC	$+0.45$	10/5	-0.39	$\mathbf{+0.24}$	11/4	$+0.20$
`no_graph`	$+0.83$	11/4	*ref*	$+0.04$	7/8	*ref*
A	$+0.70$	11/4	-0.13	$+0.03$	9/5	-0.01
BC	$+0.22$	7/8	-0.61	-1.24	0/15	-1.28
ABC	-0.53	6/9	-1.36	-0.84	1/14	-0.87
AB	-1.24	4/11	-2.07	-0.75	1/14	-0.78
B	-1.26	5/10	-2.10	-1.07	0/15	-1.10

Fourth, the BERT control behaves as a control should: every contrast is at most 0.27 MCC and none is significant, against a seed standard deviation of ≈ 1.0. Our earlier single-seed reading of this control ($45.54 \to 43.41$) was noise.

5 Three hyperbolic stages

We now turn to the paper's main question: does hyperbolic geometry extend GLOT? Stage A replaces the cosine threshold with a geodesic one in the Poincaré ball, Stage B replaces the readout with a gyro-midpoint, and Stage C replaces message passing with Möbius operations; we run all eight combinations.

Earlier campaigns of ours searched only graph-construction knobs, inheriting the learning rate, depth, width and projection width from values chosen for a *Euclidean* pooler — an asymmetry that favours the baseline and that produced a false positive we retract in Appendix C. We therefore repeat the campaign with GLOT's full grid applied identically to every arm: 40 trials per arm, 15 confirmation seeds, 990 runs. Notably *every* arm including the baseline selected $\mathrm{lr} = 2 \times 10^{-4}$; what moved was depth and width.

Stage C is the strongest extension. Hyperbolic message passing alone reaches $+1.42$ MCC on CoLA with $13/15$ seeds positive ($t = 4.11$, sign $p = 0.0074$) — significant on both tests, though above the Bonferroni threshold of 0.0063 we apply across eight arms. On STS-B, A+C is the best arm of the nine ($+0.24$, $11/4$), with C second ($+0.20$, $12/3$). Every arm containing the hyperbolic GNN *without* the readout is positive on both tasks.

The hyperbolic readout is harmful, and that is worth reporting separately. Every arm containing Stage B loses 0.75–1.24 Spearman on STS-B with 0–1 of 15 seeds positive; all four survive Bonferroni. The decoder backbone agrees (Appendix D), where the standard deviation also grows monotonically with the number of hyperbolic components ($0.48 \rightarrow 3.58$), suggesting the gyromidpoint acts as a variance amplifier rather than a useful inductive bias. Because B and C have opposite signs and similar magnitudes, a study reporting "hyperbolic pooling" as one factor would have averaged them to approximately zero and concluded that geometry does not matter here. It does; the components simply differ.

Scope: Stage C's gain is not yet separable from a no-graph control. We hold our own result to the same standard we apply elsewhere. Alongside the eight arms we ran a `no_graph` control, which also beats the cosine baseline on CoLA ($+0.83$), so part of any arm's advantage may be independent of the graph. Referenced to that control, Stage C's CoLA gain falls to $+0.58$ with 8 seeds positive and 7 negative — an exact sign-test p of 1.00. **We therefore report Stage C as the most promising direction of the three, not as an established improvement**, and we would want a density-matched Euclidean control at $n \geq 30$ before claiming more. The two arms that *do* separate from the control separate downwards (B and AB, -2.10 and -2.07 MCC, $3/12$, $p = 0.035$), and both are readout arms.

This scope condition applies to GLOT's own graph as well, and there it reads as a strength rather than a limitation: from an empty graph (`no_graph`, realised density 0.000) to a complete one (`paper_tau` on RoBERTa, 0.991), accuracy is unchanged. The same insensitivity holds for the message-passing architecture — swapping GAT for GCN or GIN moves CoLA by at most 0.52 MCC and STS-B by 0.23 (Appendix F). A method this robust to its own hyper-parameters is unusually easy to deploy on a new backbone. GLOT does outperform mean, max and `[CLS]` pooling — by 14–28 points on the authors' own noise diagnostic, in every seed — but a GLOT with *zero* message-passing layers matches the full model there at every noise level, and so does a trainable attention pooler with no graph at all (Appendix E). Taken together, our reading is that much of the benefit is carried by the trained readout, with the graph acting as a token-selection mechanism whose exact construction is not critical. We regard that as a useful localisation for future work on this family, including our own: it says the readout is the component most likely to repay further design effort, which is consistent with Stage C helping and Stage B hurting. Appendix D shows the same ordering on a decoder backbone.

A third task. MRPC, run to the same standard (wide search, $n = 15$ paired seeds, minimum detectable effect 0.317 F1), reproduces the ordering without adding a positive: no arm beats the baseline, `no_graph` is indistinguishable from it ($+0.026$, $9/6$), Stage A is likewise flat ($+0.040$, $8/7$), and every hyperbolic arm is negative (-0.151 to -0.333), none significantly. Stage C's CoLA advantage therefore does not extend to MRPC, which we note as a limit on its generality.

Scope. These results are BERT on CoLA, STS-B and MRPC; RTE is still in progress. Two implementation details in the released code affect anyone reproducing this grid, and we document them in Appendix H so that others do not lose time to them.

6 CONCLUSION

We set out to ask whether hyperbolic geometry extends GLOT, and the answer depends entirely on which component receives it. Hyperbolic message passing (Stage C) is the most promising of the three, giving $+1.42$ MCC on CoLA with $13/15$ seeds positive and pairing with Stage A to produce the best STS-B arm we measured; the hyperbolic readout (Stage B) is significantly harmful on both an encoder and a decoder backbone. Had we reported "hyperbolic pooling" as one factor, these would have cancelled. Stage C's gain is not yet separable from a no-graph control, so we present it as a direction rather than a result, and we say what would settle it: a density-matched Euclidean control at $n \geq 30$.

Two conclusions about GLOT follow, both favourable. It performs better than its own paper reports, by 3.03 MCC on CoLA and 0.36 on RTE at the authors' seed over the authors' grid, because published numbers are single-seed maxima rather than seed means — a reporting convention, not a modelling issue. And it is strikingly insensitive to the graph it is given: realised density spans 0.00 to 0.99 across backbones and arms with no change in accuracy, and the GNN backbone can be swapped without effect. For a method intended to be dropped on top of an arbitrary frozen encoder, that robustness is a genuine practical asset, and our measurements let a practitioner exploit it: port GLOT as published rather than re-tuning the threshold, and avoid mean-based norm rescaling on backbones with attention-sink outliers, which costs 11 MCC.

Finally, a methodological note that applies to us as much as to anyone: at these effect sizes the difference between a seed mean and a grid maximum is larger than most of the differences being claimed, so the estimator should be reported alongside the number. Two of our own apparently positive results did not survive that discipline, and we report them as retractions in Appendices C and G.

References

Krishna Sri Ipsit Mantri, Carola-Bibiane Schönlieb, Zorah Lähner, and Moshe Eliasof. Towards improved sentence representations using token graphs. In *International Conference on Learning Representations (ICLR)*, 2026. Verified against extracted PDF text; add URL/pages before submission.

A Experimental protocol

Frozen backbone, 2 epochs, Adam, learning rate 2×10^{-4}, weight decay 0, batch size 32; the pooler is the only trained component. Confirmation uses the fixed seed set $\{1, \ldots, 15\}$ shared across every arm; tuning uses a single disjoint held-out seed.

Matched budget. Every arm receives 40 trials in the wide campaign (10 in the graph-only campaigns). The cosine baseline has a smaller native search space, so we expand its grid to keep selection bias equal. Equal trials is not equal *coverage*: 40 trials cover 6.2% of `no_graph`'s 648-point space but under 0.01% of ABC's 4.98×10^7-point space. This asymmetry disfavours the complex arms, and we note it as a caveat against our own negative results rather than in support of them.

Testing. We report the exact two-sided sign test alongside the paired t-test because evaluation-set quantisation makes the parametric standard error misleadingly small (MRPC accuracy moves in steps of 0.245 on 408 examples). A result counts as significant only when both tests agree. Bonferroni over the 8 arms tested against a common baseline gives $\alpha = 0.0063$.

Cache/RNG confound. The released code precomputes and caches backbone hidden states, and constructs the shuffled `DataLoader` *before* initialising the classifier. On a cache miss the loader is iterated and consumes `torch.randperm` from the global RNG; on a hit it is not. The same seed therefore yields different classifier initialisation depending on cache state — measured at 40.37 (cold) versus 45.54 (warm) MCC on CoLA, about $6\times$ the seed standard deviation. All caches are built before any arm runs. Any reproduction that skips this will shift by several points on CoLA.

B Full arm statistics

Table 5 reports point estimates and seed counts. The corresponding paired statistics on STS-B are: B $\bar{d} = -1.065$, $t = -17.00$, sign $p = 6 \times 10^{-5}$; BC -1.242, $t = -8.30$, $p = 6 \times 10^{-5}$; ABC -0.837, $t = -6.36$, $p = 0.00098$; AB -0.747, $t = -5.66$, $p = 6 \times 10^{-5}$. All four survive Bonferroni. On CoLA, C gives $\bar{d} = +1.416$, $t = 4.11$, sign $p = 0.0074$ — significant on both tests but above $\alpha = 0.0063$. The minimum detectable effect at $n = 15$ is 0.225 (STS-B) and 1.153 (CoLA).

C A RETRACTED RESULT OF OUR OWN, AND WHY

Under the graph-only search, Stage A on STS-B gave $\bar{d} = +0.223$, $t(14) = 6.20$, sign $p = 6\times10^{-5}$, $15/15$ seeds positive, and replicated on 12 held-out seeds. It does not survive the wide search: the same comparison gives $\bar{d} = +0.028$, $t = 0.69$, $9/5$ seeds. The effect was real but its cause was not geometry — the baseline had been pinned at $K = 2$, $h = 128$ while nothing else was, and it recovers the entire gap once it can choose $K = 4$. We report this because it is the clearest illustration of the paper's own point: a pooling comparison is only as trustworthy as the budget given to its control.

D DECODER BACKBONE

TinyLlama-1.1B on STS-B, $n = 15$ shared seeds. This backbone was selected over SmolLM2 on measured geometry: Table 2 puts it in BERT's regime (0.073 against BERT's 0.103 at $\tau = 0.6$) whereas SmolLM2 sits at 0.741. The minimum detectable effect is 0.503.

Densities below are recovered from training telemetry, which uses the defective statistic described in Appendix H: it reports $(|E_{\text{off}}| + n)/(n(n-1))$, so an empty graph reads $1/(n-1)$ rather than 0. We subtract that offset, which the `no_graph` arm identifies exactly ($0.095 \to 0$) and which is internally consistent across all seven arms. These are STS-B sentences and so are not directly comparable with the CoLA measurements of Table 2.

Arm	mean	std	density	Δ	sign p	pos/neg
AC	80.15	0.54	0.027	+0.205	0.302	10/5
A	79.99	0.51	0.099	+0.043	0.607	9/6
`no_graph`	79.95	0.48	0.000	+0.001	1.000	8/7
baseline	79.95	0.49	0.026	*ref*	—	—
BC	78.78	0.95	0.101	−1.165	0.00098	1/14
AB	76.39	1.87	0.100	−3.556	6×10^{-5}	0/15
ABC	75.44	3.58	0.100	−4.506	6×10^{-5}	0/15

The ordering matches BERT: `no_graph` is indistinguishable from the baseline ($\Delta = +0.001$, $p = 1.00$) despite the baseline having a well-conditioned graph, and the readout arms are significantly harmful. Standard deviation grows monotonically with the number of hyperbolic components ($0.48 \to 0.95 \to 1.87 \to 3.58$), consistent with these operators acting as variance amplifiers rather than as useful inductive bias. Realised density spans 0.000–0.101 across arms, so a density-matched Euclidean control is still needed to fully separate sparsity from geometry; this matters least for the two arms that matter most, since AC sits at 0.122 against the baseline's 0.121.

E DIAGNOSTIC STRESS TEST

We reproduce the signal-in-noise diagnostic of Mantri et al. (2026): a short signal phrase carrying a logical dependency, injected into a sequence of random distractor words, with the distractor ratio swept from 20% to 90%. BERT backbone, mean $\pm$ std over 5 seeds; the original reports a single seed.

Arm	20%	50%	80%	90%
`no_graph`	**98.0** ±1.3	**94.8** ±5.7	**92.5** ±5.0	84.4 ±10.1
AC	95.0 ±3.4	88.2 ±11.2	90.8 ±5.8	**88.5** ±8.9
C	95.0 ±3.3	87.6 ±11.2	90.8 ±5.9	88.4 ±8.8
A	97.0 ±1.3	91.9 ±4.1	91.0 ±2.7	84.9 ±9.1
baseline	96.0 ±0.5	90.2 ±5.0	88.1 ±4.7	79.8 ±10.8
B	95.4 ±1.4	88.7 ±2.8	83.8 ±13.8	75.5 ±14.9
Mantri et al. (2026) Table 7, BERT row (single seed):				
GLOT	97.2	97.0	97.8	98.8

Two observations. The ordering again places `no_graph` at or above the baseline at every ratio, in the experiment designed specifically to demonstrate the graph's value.

Second, we do not reproduce the published GLOT numbers: at 90% distractors the original reports 98.8 where we measure 79.8 ± 10.8 for the same method, and their curve is flat or rising under increasing noise where ours degrades monotonically. We flag this as unresolved rather than as a refutation — the generation procedure is underspecified in the original (vocabulary source and signal-phrase templates are not given), so the gap is plausibly task construction rather than method.

Competing poolers. Mantri et al. (2026)'s claim is comparative: GLOT holds up under noise where other poolers collapse. We ran those poolers on our own generator, 5 shared seeds each, together with a GLOT variant at $K = 0$ that keeps the adaptive scorer but performs no message passing — the parameter-matched control for whether the benefit is relational or simply capacity. Deltas below are paired against full GLOT on the same seeds, which is a comparison internal to our data and so does not depend on matching the published values.

Pooler	type	20%	50%	80%	90%
GLOT (K=0)	trained	95.7	91.0	89.3	88.1
AdaPool	trained	95.1	90.8	86.5	83.6
GLOT	trained	96.0	90.2	88.2	79.7
Mean	static	73.5	63.2	64.0	57.4
`[CLS]`	static	70.0	61.4	62.1	65.4
Max	static	63.0	55.2	53.1	51.8
paired Δ vs GLOT *(pos/neg of 5 seeds, exact sign test):*					
K=0		-0.3 (2/3, 1.00)	$+0.8$ (2/3, 1.00)	$+1.0$ (4/1, 0.38)	$+8.4$ (3/2, 1.00)
AdaPool		-0.8 (1/4, 0.38)	$+0.7$ (3/1, 0.63)	-1.8 (2/3, 1.00)	$+3.9$ (4/1, 0.38)
static		-14 to -35, $0/5$ at every ratio ($p = 0.062$, the floor at $n = 5$)			

Two conclusions, and one caution about what this table cannot support.

The trained/static split is unambiguous and holds in every seed: mean, `[CLS]` and max lose 14–35 points to GLOT at every ratio, $0/5$ seeds positive. GLOT's advantage over set-based pooling is real.

Within the trained group, however, nothing separates. $K = 0$ — no message passing, no graph consumed at all — is statistically indistinguishable from full GLOT at every ratio, and so is AdaPool, which has no token graph whatsoever. We emphasise that we do *not* claim $K = 0$ is better: its $+8.4$ at 90% rests on $3/5$ seeds with a sign-test p of 1.00 and per-seed sd of 5.8 and 9.7. The claim is the null one, and it is the relevant one: the diagnostic designed to demonstrate that relational reasoning over the token graph confers noise robustness does not distinguish a model that performs that reasoning from two models that do not.

Caution: our trained poolers do not match theirs. The divergence between our numbers and the published ones is not uniform, and the pattern matters. The static poolers agree closely (-2.2 to $+5.6$ across all twelve cells). Both *trained* poolers diverge instead, in opposite directions, and the divergence grows with noise: at 90% our AdaPool is 22.0 points *above* the published value while our GLOT is 19.1 points *below* it. Consequently the 37-point margin their Table 7 reports for GLOT over AdaPool at 90% becomes -3.9 in our hands. We therefore cannot claim to reproduce their setup for trained poolers, and we do not use the published values as a baseline anywhere; only the paired within-data deltas above carry weight. The comparison their claim rests on is exactly the one that fails to replicate, and their Appendix B.4 does not specify the vocabulary source, the signal-phrase templates, or the pooler training budget, so we cannot localise the cause.

F GNN BACKBONE AND TRAINING RECIPE

GNN backbone. Every arm in the main campaigns pins `gnn_type=gat`, so a reviewer may reasonably ask whether "the graph carries no information" is really "GAT extracts none". It is not: over 5 seeds with everything else at the published recipe, CoLA gives GAT 45.38 ± 1.19, GCN 45.14 ± 1.78, GIN 44.86 ± 1.33, and STS-B gives 82.69 ± 0.34, 82.46 ± 0.27, 82.47 ± 0.32. The spread is 0.52 MCC and 0.23 Spearman, well inside the seed noise, and GAT is the only one of the three that consumes edge attributes at all. Message-passing architecture is not where the variation lives either.

The STS-B residual is not a recipe mismatch. §3 leaves an unexplained -0.87 on STS-B. The released README trains differently from the paper's Appendix B.2 (3 epochs, `scorer_hidden` 256, $\tau = 0.8$ versus 2, 128, 0.6), so we ran the $2 \times 2 \times 2$ decomposition at the authors' seed 42. Epoch count is the only factor that moves anything ($82.36 \rightarrow 82.66$ for $2 \rightarrow 3$ epochs); τ and `scorer_hidden` are inert to within 0.04. Over 5 seeds the two full recipes are indistinguishable (82.69 ± 0.34 against 82.69 ± 0.23). The best cell we obtain, 82.66, still sits 1.20 below the published 83.86, so the residual is not explained by the recipe discrepancy and we leave it open.

G Measurement artifacts

δ-hyperbolicity on raw activations measures attention sinks. Motivating hyperbolic pooling by measured tree-likeness is misleading on raw activations. ModernBERT at layer 16 has $\delta_{\text{eucl}} = 0.0021$, which would indicate an almost perfectly tree-like space; its ratio of maximum to median token norm is 156. Computing δ on the *angular* metric $\arccos(\cos(\cdot, \cdot))$, which is invariant to those outliers, gives 0.2127 — statistically indistinguishable from BERT's 0.2114. Flan-T5 shows the same pattern with $357\times$ norm outliers. We recommend reporting angular δ alongside Euclidean δ together with the norm-outlier ratio in any work motivated by measured hyperbolicity.

Tuning maxima are inflated. With T trials and per-seed noise σ, the expected maximum is inflated even for an arm identical to the baseline: ≈ 1.83 here (STS-B/BERT, $\sigma = 1.19$) and ≈ 0.83 (STS-B/TinyLlama, $\sigma = 0.54$). Several apparent leads in this project were smaller than this and did not survive confirmation. Relatedly, the noise floor must be measured for the configuration being confirmed: we initially assumed CoLA's $\sigma = 0.81$ from a layer-12 probe, where the true value at layer 8 is 2.09.

The RoBERTa campaign gives the sharpest illustration, because the inflation there is large enough to *reverse the ranking*. Mean drop from best-tuning score to the 15-seed confirmation mean is 2.08 MCC on CoLA and 0.48 Spearman on STS-B, and no arm's rank is preserved: `density_fix` is first of five at the tuning maximum (56.77) and last of five over 15 seeds (53.19). Any protocol that reports the tuning maximum would have concluded the opposite of what we report in §4.

Layer selection overfits. Selecting a backbone layer on one task does not transfer: layer 8 beat layer 12 by $+4.62$ MCC on CoLA but by only $+0.25$ on STS-B, and lost on RTE (-0.36) and MRPC (-0.46).

H Notes for reproducers

Three details of the released implementation cost us time and are recorded here so they do not cost anyone else's. None affects the conclusions of the original paper.

- `jk_mode=lstm` is accepted by the argument parser but raises at runtime, and the advertised options `mean` and `none` are not implemented. The effective space is $\{$`cat`, `max`$\}$.
- Thresholding Poincaré *distance* at an absolute radius yields an empty graph on real features. BERT token norms (≈ 14.7) saturate $\exp_0^c$ to radius 1.0 exactly in `float32`, and real pairwise geodesic distances land in $[8.85, 11.76]$, so any absolute threshold below ≈ 9 gives zero edges. The diagnostic signature is a score that is bit-identical across an entire threshold grid. We use a quantile threshold throughout for this reason.
- The logged edge-density statistic counts self-loops in its numerator but excludes the diagonal from its denominator, so it reports $|E|/(n(n-1))$ with $|E|$ including the n self-loops. A complete graph therefore logs $n/(n-1) > 1$ and an *empty* graph logs $1/(n-1) \approx 0.10$ at CoLA's typical length rather than 0. This is a reporting defect only — no training path consumes the statistic, so no score is affected — but it is a trap for exactly the audit this paper performs. Two consequences are worth recording. Our `no_graph` control logs density 0.108 on BERT/CoLA, which looks like a live graph and is in fact the empty-graph signature $1/(n-1)$; and the corrected BERT baseline density, $0.2074 - 0.10 \approx 0.104$, independently reproduces the 0.103 measured directly in Table 2, which is what confirms the correction. Any density we quote is re-measured with the standalone instrument described in Table 2, never taken from training telemetry.